Homomorphism form of Polynomial Freiman-Ruzsa

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github:teorth-pfr:4c56de547b2f33b0bc47b92f3bfbcebc6457a776;
Apache-2.0

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Editorial note. Ordinary cited prerequisites and concise author arguments are accepted as such. No independent mathematical certification or public ID allocation is claimed.

Allowed standard prerequisites (editorial).

- Standard finite Shannon entropy identities, conditional entropy and finite abelian group algebra are cited in their original Blueprint formulations.
- Named prior PFR/entropy or BSG results are allowed prerequisites; their original statements are retained where used, without demanding recursively expanded proofs.

1 Proof of PFR

Lemma 1 (Ruzsa covering lemma)**.** *If A, B are finite non-empty subsets of a group G, then A can be covered by at most $|A+B|/|B|$ translates of $B-B$.*

Proof. Cover A greedily by disjoint translates of B. □

Proposition 2. *For any random variables Y_1, Y_2, there exist a subgroup H such that*

$$2\rho(U_H) \leq \rho(Y_1) + \rho(Y_2) + 8d[Y_1; Y_2].$$

Corollary 3. *If $|A+A| \leq K|A|$, then there exists a subgroup H and $t \in G$ such that $|A \cap (H+t)| \geq K^{-4}\sqrt{|A||H|}$, and $|H|/|A| \in [K^{-8}, K^8]$.*

Corollary 4. *If $|A+A| \leq K|A|$, then there exist a subgroup H and a subset c of G with $A \subseteq c+H$, such that $|c| \leq K^5|A|^{1/2}/|H|^{1/2}$ and $|H|/|A| \in [K^{-8}, K^8]$.*

2 Homomorphism version of PFR

In this section, G, G' are finite abelian 2-groups.

Lemma 5 (Hahn-Banach type theorem). *Let H_0 be a subgroup of G. Then every homomorphism $\phi : H_0 \to G'$ can be extended to a homomorphism $\tilde{\phi} : G \to G'$.*

Proof. By induction it suffices to treat the case where H_0 has index 2 in G, but then the extension can be constructed by hand. □

Lemma 6 (Goursat type theorem). *Let H be a subgroup of $G \times G'$. Then there exists a subgroup H_0 of G, a subgroup H_1 of G', and a homomorphism $\phi : G \to G'$ such that*

$$H := \{(x, \phi(x) + y) : x \in H_0, y \in H_1\}.$$

In particular, $|H| = |H_0||H_1|$.

Proof. We can take H_0 to be the projection of H to G, and H_1 to be the slice $H_1 := \{y : (0, y) \in H\}$. One can construct ϕ on H_0 one generator at a time by the greedy algorithm, and then extend to G by Theorem 5. The cardinality bound is clear from direct counting. □

Theorem 7 (Homomorphism form of PFR). *Let $f : G \to G'$ be a function, and let S denote the set*

$$S := \{f(x + y) - f(x) - f(y) : x, y \in G\}.$$

Then there exists a homomorphism $\phi : G \to G'$ such that

$$|\{f(x) - \phi(x) : x \in G\}| \leq |S|^{10}.$$

Proof. Consider the graph $A \subset G \times G'$ defined by

$$A := \{(x, f(x)) : x \in G\}.$$

Clearly, $|A| = |G|$. By hypothesis, we have

$$A + A \subset \{(x, f(x) + s) : x \in G, s \in S\}$$

and hence $|A + A| \leq |S||A|$. Applying Theorem 4, we may find a subspace $H \subset G \times G'$ such that $|H|/|A| \in [|S|^{-8}, |S|^8]$ and A is covered by $c + H$ with $|c| \leq |S|^5 |A|^{1/2}/|H|^{1/2}$. If we let H_0, H_1 be as in Theorem 6, this implies on taking projections that G is covered by at most $|c|$ translates of H_0. This implies that

$$|c||H_0| \geq |G|;$$

since $|H_0||H_1| = |H|$, we conclude that

$$|H_1| \leq |c||H|/|G| = |c||H|/|A|.$$

By hypothesis, A is covered by at most $|c|$ translates of H, and hence by at most $|c||H_1|$ translates of $\{(x, \phi(x)) : x \in G\}$. As ϕ is a homomorphism, each such translate can be written in the form $\{(x, \phi(x) + d) : x \in G\}$ for some $d \in G'$. Since

$$|c||H_1| \leq |c|^2 \frac{|H|}{|A|} \leq \left(|S|^5 \frac{|A|^{1/2}}{|H|^{1/2}}\right)^2 \frac{|H|}{|A|} = |S|^{10},$$

the result follows. □